

1 Relations And Functions

Relation is a set of ordered pairs that can be represented by a diagram, graph, or sentence.

The **Domain** is the set of all *first* numbers of the ordered pairs in a relation.

The **Range** is the set of all *second* numbers of the ordered pairs in a relation.

A **Function** is when none of the domain values are repeated.

A **Vertical Line Test** is a test for functions.

1.1 Compositions of Functions

A **composite function** is when a variable is replaced with another function.

Example: $f(g(x))$ is read as: 'f of g of x'

Inverse Relations is when you reverse ordered pairs of an initial relation.

The **identity function** is the line which if you folded the graph over on this line/crease it would line up both relations. Since for each replacement of x, the result is identical to x.

